

The Language of Set Theory

Robert Y. Lewis

CS 0220 2021

February 9, 2022

Overview

- 1 Sets Definitions (4.1–4.1.1)
- 2 Sets Operations (4.1.2–4.1.5)
- 3 Equality Proofs (4.1.6)
- 4 Venn Diagrams

The language of sets

We're building up a new vocabulary.

Are we *defining* sets in terms of something else? Or *introducing* a new concept?

Set Definition

Definition (informal): A *set* is a bunch/collection/group of objects.

Definition: The *elements* of the set are the objects contained in that set.

Sets can contain numbers, ordered sequences of numbers, strings, names, or other sets.

Objects are either *in* the set or *not in* the set. We don't have a concept of an object being in a set multiple times. It's a Boolean property.

We write curly braces around a comma-separated list to build a set.

Examples:

- $H = \{ \text{Jiahua, Nick, Rachel, Walter} \}$
- $I = \{ \text{this laptop, this slide clicker, that projector screen} \}$
- $J = \{ \text{"this laptop", "this slide clicker", "that projector screen"} \}$
- $\mathbb{N} = \{0, 1, 2, 3, 4, \dots\}$

Elements

- $H = \{ \text{Jiahua, Nick, Rachel, Walter} \}$
- $I = \{ \text{this laptop, this slide clicker, that projector screen} \}$
- $\mathbb{N} = \{0, 1, 2, 3, 4, \dots\}$

Definition: We say $x \in S$ if x is *an element of* or *in* or *a member of* the set S .

- $\text{Jiahua} \in H$? Yes.
- $\text{this laptop} \in H$? No. This laptop $\notin H$.
- $\text{this laptop} \in I$? Yes.
- $\text{Rachel} \in \mathbb{N}$? No.

Some Sets of Numbers

- $\emptyset = \{\}$ (empty set, null set)
- $\mathbb{N} = \{0, 1, 2, 3, 4, \dots\}$ (non-negative integers)
- $\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$ (integers)
- $\mathbb{Q} = \{1/2, -4/15, 21, \dots\}$ (rationals)
- $\mathbb{R} = \{\sqrt{2}, -\pi, 21, \dots\}$ (real numbers)
- $\mathbb{C} = \{i/2, 15 - i, \sqrt{7}, 21, \dots\}$ (complex numbers)

Superscript plus limits to (strictly!) positive values: $\mathbb{Z}^+ = \mathbb{N}^+$.

Superscript minus limits to negative values: $21 \notin \mathbb{R}^-$.

Sets of sets

■ $A = \{1, 4, 9\}$

■ $B = \{\{1, \{4\}\}, \{9\}\}$

■ $1 \in A$? Yes.

■ $1 \in B$? No, but $\{1, \{4\}\} \in B$.

■ $\exists x \in B, 1 \in x$? Yes, $x = \{1, \{4\}\} \in B$ and $1 \in x$.

Subsets

Definition: One set is a *subset* of another if every element of the first set is also an element of the second.

We write $S \subseteq T$ to say the set S is a subset of set T . So, $S \subseteq T$ means $\forall x \in S, x \in T$. Could also write $\forall x, x \in S \rightarrow x \in T$.

Examples:

- $\mathbb{N} \subseteq \mathbb{Z}$? Yes, every positive integer is also an integer.
- $\mathbb{Z}^+ \subseteq \mathbb{N}$? Yes, every positive integer is also a non-negative integer.
- $\mathbb{C} \subseteq \mathbb{Z}$? No, $\mathbb{C} \not\subseteq \mathbb{Z}$. Some (many!) complex numbers are not integers. Although, $\mathbb{Z} \subseteq \mathbb{C}$.
- $\mathbb{N} \subseteq \mathbb{N}$. Yes, if sets are equal, all of the first must also be in the second!

Note: $\{1, 2, 3\} \subseteq \{1, 2, 3, 4\}$ looks a little bit like $3 \leq 4$.

We write $A \subset B$ to rule out equality (like $a < b$).

Non-Trichotomy

If a and b are integers, exactly one of these properties must hold:

■ $a < b$

■ $a = b$

■ $a > b$

Not so for subsets. Example?

$$A = \{0\}$$

$$B = \{1\}$$

$A \subset B$? No, $0 \in A$, but $0 \notin B$.

$A = B$? No, they are different sets.

$A \supset B$? (superset!) No, $1 \in B$, but $1 \notin A$.

Operations on sets: Union

$$\blacksquare A = \{j, i, a, h, u\}$$

$$\blacksquare B = \{n, i, c, k\}$$

$$\blacksquare C = \{r, a, c, h, e, l\}$$

$$\blacksquare D = \{w, a, l, t, e, r\}$$

Definition: The *union* of sets X and Y , $X \cup Y$, consists of every element that is in either X or Y . In other words, $z \in X \cup Y$ means $z \in X \vee z \in Y$.

Example: $A \cup B = \{j, i, a, h, u, n, c, k\}$. (Order doesn't matter: $= \{a, c, h, i, j, k, n, u\}$)

Operations on sets: Intersection

$$\blacksquare A = \{j, i, a, h, u\}$$

$$\blacksquare B = \{n, i, c, k\}$$

$$\blacksquare C = \{r, a, c, h, e, l\}$$

$$\blacksquare D = \{w, a, l, t, e, r\}$$

Definition: The *intersection* of sets X and Y , $X \cap Y$, consists of every element that is in both X and Y . In other words, $z \in X \cap Y$ means $z \in X \wedge z \in Y$.

Example: $C \cap D = \{r, a, e, l\}$. $B \cap D = \emptyset$

Operations on sets: Set difference

■ $A = \{j, i, a, h, u\}$

■ $B = \{n, i, c, k\}$

■ $C = \{r, a, c, h, e, l\}$

■ $D = \{w, a, l, t, e, r\}$

Definition: The *set difference* of sets X and Y , $X \setminus Y$, consists of every element that is in X but not in Y . In other words, $z \in X \setminus Y$ means $z \in X \wedge z \notin Y$.

Example: $C \setminus D = \{c, h\}$.

Example: $B \setminus A = \{n, c, k\}$.

Operations on sets: Symmetric difference

$$\blacksquare A = \{j, i, a, h, u\}$$

$$\blacksquare B = \{n, i, c, k\}$$

$$\blacksquare C = \{r, a, c, h, e, l\}$$

$$\blacksquare D = \{w, a, l, t, e, r\}$$

Definition: The *symmetric difference* of sets X and Y , $X \triangle Y$, consists of every element that is in X but not in Y or in Y but not X . In other words, $z \in X \triangle Y$ means $(z \in X \wedge z \notin Y) \vee (z \in Y \wedge z \notin X)$. That is, $z \in X \text{ XOR } z \in Y$.

Example: $C \triangle D = \{c, h, w, t\}$.

Example: $A \triangle C = \{r, c, e, l, j, i, u\}$. (Remember: order doesn't matter)

Operations on sets: Complement

■ $A = \{j, i, a, h, u\}$

■ $B = \{n, i, c, k\}$

■ $C = \{r, a, c, h, e, l\}$

■ $D = \{w, a, l, t, e, r\}$

Definition: The *complement* of a set X , $\bar{X}$, is defined with respect to some universe of possible elements U . It consists of every possible element that is not in X . In other words, $\bar{X} = U \setminus X$.

Example: If U is the universe of all letters in English, $\bar{B} = \{a, b, d, e, f, g, h, j, l, m, o, p, q, r, s, t, u, v, w, x, y, z\}$.

Example: If $U = \mathbb{Z}$, $\mathbb{Z}^- = \overline{\mathbb{Z}^+} \setminus \{0\}$.

Disjoint sets

Definition: Sets A and B are *disjoint* if they have no elements in common.

$$A \cap B = \emptyset \text{ or}$$

$$A \subseteq \overline{B}.$$

Operations on sets: Power set

$$\blacksquare A = \{j, i, a, h, u\}$$

$$\blacksquare B = \{n, i, c, k\}$$

$$\blacksquare C = \{r, a, c, h, e, l\}$$

$$\blacksquare D = \{w, a, l, t, e, r\}$$

Definition: The *power set* of a set X , $\mathcal{P}(X)$, is the set of all subsets of X . In other words, $\forall x \in \mathcal{P}(X), x \subseteq X$ and $\forall x \subseteq X, x \in \mathcal{P}(X)$.

Example: If $E = A \cap C = \{a, h\}$, $\mathcal{P}(E) = \{\{\}, \{a\}, \{h\}, \{a, h\}\}$.

Example: $\mathcal{P}(B) = \{\{\}, \{n\}, \{i\}, \{c\}, \{k\}, \{n, i\}, \{n, c\}, \{n, k\}, \{i, c\}, \dots, \{n, i, c, k\}\}$.

Example: $\mathcal{P}(\emptyset) = \{\emptyset\}$.

Products of sets

■ $C = \{2, 5\}$

■ $D = \{a, b, c\}$

■ $C \times D = \{(2, a), (2, b), (2, c), (5, a), (5, b), (5, c)\}$

■ $\mathbb{N} \times D = \{(0, a), (0, b), (0, c), (1, a), (1, b), \dots\}$

■ $\mathbb{N} \times \mathbb{N} =$ the set of *ordered pairs* of natural numbers

Ordered pair: $(2, 0)$ is not the same as $(0, 2)$!

Operations on sets: Cardinality

■ $A = \{j, i, a, h, u\}$

■ $B = \{n, i, c, k\}$

■ $C = \{r, a, c, h, e, l\}$

■ $D = \{w, a, l, t, e, r\}$

Definition: The *cardinality* of a set X , $|X|$, is the count of the number of (unique) elements in X .

Example: $|A| = 5$, $|B| = 4$, $|C| = 6$, $|D| = 6$.

Example: $|\emptyset| = 0$.

Example: If $|A| = n$, $|\mathcal{P}(A)| = 2^n$. Each subset consists of a decision of whether to include or not include (2 possibilities) each of the n elements of A .

Building sets with predicates

General form: $\{ \text{description of a set} \mid \text{filter on the set} \}$.

Examples:

■ $A = \{n \in \mathbb{N} \mid n = 2k + 1 \text{ for some integer } k\}$

■ $B = \{x \in \mathbb{R} \mid x^2 > 1\}$

Note: Python has a notation for this idea.

Proving set equalities: Logic

$A = B$ means that, for all x , $x \in A \leftrightarrow x \in B$. We can use this definition to prove various set equalities. Here's a useful one.

Theorem: $A = B$ if and only if both $A \subseteq B$ and $B \subseteq A$.

$A = B$ is equivalent to

$\forall x, x \in A \leftrightarrow x \in B$ which is equivalent to

$\forall x, ((x \in A \rightarrow x \in B) \wedge (x \in B \rightarrow x \in A))$ which is equivalent to

$(\forall x, (x \in A \rightarrow x \in B)) \wedge (\forall x, (x \in B \rightarrow x \in A))$ which is equivalent to

$\forall x \in A, x \in B \wedge \forall x \in B, x \in A$ which is equivalent to

$A \subseteq B \wedge B \subseteq A$.

Proving set equalities: set-element method

We can use the previous result to prove other set equalities.

Theorem: For any sets A and B of elements in universe U , $\overline{A \cap B} = \bar{A} \cup \bar{B}$.

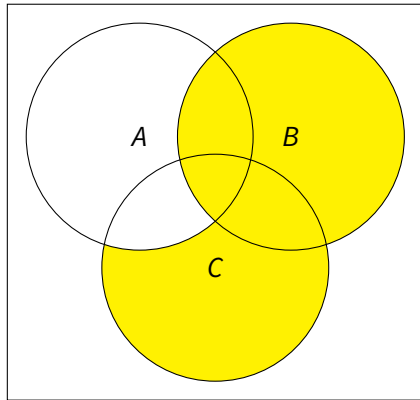
DeMorgan's Law again! Now, connects intersection and union instead of $\wedge$ and $\vee$.

An object $x \in \overline{A \cap B}$ if it is *not* in both A and B . Such an element must either not be in A or not be in B . It follows that such an element must be in $\bar{A} \cup \bar{B}$. Thus, we have $\overline{A \cap B} \subseteq \bar{A} \cup \bar{B}$.

Note also that an object $x \in \bar{A} \cup \bar{B}$ if it is either not in A or it is not in B . Such an element can't be in both A and B , therefore. Said another way, it must be in $\overline{A \cap B}$. Thus, we have $\bar{A} \cup \bar{B} \subseteq \overline{A \cap B}$.

Since both $\overline{A \cap B} \subseteq \bar{A} \cup \bar{B}$ and $\bar{A} \cup \bar{B} \subseteq \overline{A \cap B}$ are true, we know $\overline{A \cap B} = \bar{A} \cup \bar{B}$.

Venn diagram translation



$$(C \setminus A) \cup B$$

$$(C \setminus A \setminus B) \cup ((C \cap B) \setminus A) \cup (A \cap B \cap C) \cup ((A \cap B) \setminus C) \cup (B \setminus A \setminus C)$$